AI AUTOMATION SPEND IN BANKING SECTOR ANALYSIS

A Data Analytics Study of Investment, Cost Savings, ROI and Automation Impact

Research Paper

Dataset Scope	**72 banking AI/automation records**
Analysis Period	2020–2025
Countries	12
Regions	4
Banks	27
AI Use Cases	10
Technologies	4

Abstract

This paper presents a data analytics study of artificial intelligence and automation spending across banking organizations. The project examines investment levels, estimated cost savings, return on investment, automation rates, monthly FTE hours saved, AI project adoption and maturity. The analysis uses a 72-record banking dataset covering 2020 to 2025, with records from 12 countries, 27 banks, four regions, ten AI use cases and four technology categories. Python, Pandas and NumPy were used for data preparation and feature engineering, while Matplotlib and Seaborn supported exploratory analysis and visualization. SQL/MySQL was included for structured database analysis. The study derives net benefit, savings-to-spend ratio and a composite total impact score to extend the business interpretation of the raw measures. Across the dataset, total AI spending is USD 547.26 million against estimated cost savings of USD 569.51 million, producing a net benefit of USD 22.25 million. The average ROI is 1.96x, the average automation rate is 61.57%, and the records represent 5,099 AI projects with 34,684 monthly FTE hours saved. The findings provide a structured view of how banking organizations are allocating AI and automation investment and how those investments relate to operational outcomes.

Keywords: Artificial Intelligence, Banking, Automation, AI Spending, ROI, Cost Savings, Data Analytics, Python, SQL, Machine Learning, Generative AI

1. Introduction

Artificial intelligence and automation are increasingly used by banking organizations to improve operational efficiency, manage risk, process documents, support customers and modernize technology operations. Banking is a data-intensive industry, so AI applications can influence both the cost structure and the speed of business processes.

The project examined in this paper focuses on the economic and operational side of AI adoption rather than treating AI only as a technology topic. The dataset records AI spending together with estimated cost savings, ROI, automation rate, FTE hours saved, number of AI projects and maturity. This makes it possible to compare investment with measurable business-oriented indicators.

The study follows a practical analytics workflow: raw data inspection, cleaning, validation, feature engineering, exploratory data analysis, relationship analysis, SQL-based aggregation and generation of business insights. The project artifacts also contain a MySQL database design and a Python-to-MySQL connection script.

1.1 Objectives

- Measure the total and year-wise AI spending recorded in the dataset.
- Compare estimated cost savings with AI investment and calculate net benefit.
- Study ROI, automation rate and FTE hours saved across banking AI initiatives.
- Compare AI investment across countries, regions, banks, technologies and use cases.
- Use feature engineering to create business-oriented metrics from the original fields.
- Use SQL and Python-based analysis to support structured and exploratory reporting.

1.2 Research Questions

- How has recorded AI spending changed between 2020 and 2025?
- Which regions and countries account for the largest recorded AI spending?
- Which technologies and use cases are most prominent in the dataset?
- How do AI spending, cost savings, ROI and automation relate to one another?
- What operational impact is represented by monthly FTE hours saved?

2. Dataset and Methodology

The project package contains a raw CSV, a cleaned and feature-engineered CSV, two Jupyter notebooks, a SQL script and a Python database connection file. The raw dataset contains 72 records and 13 original fields. After processing, the analytical dataset contains 16 fields because three derived business metrics were added.

The core fields include Year, Region, Country, Bank, AI Use Case, Technology, AI Spend, Estimated Cost Savings, ROI Multiple, Automation Rate, FTE Hours Saved Monthly, AI Projects and Maturity. The cleaned dataset additionally contains Net Benefit, Savings-to-Spend Ratio and Total Impact Score.

2.1 Data Cleaning

- Column names were standardized by removing extra spaces and non-alphanumeric characters.
- Duplicate rows were checked and removed during preprocessing.
- Text fields were stripped of unnecessary leading and trailing spaces.
- Numeric columns were converted using numeric coercion to validate data types.
- Post-cleaning missing-value and duplicate checks were performed.

2.2 Feature Engineering

Net Benefit = Estimated Cost Savings − AI Spend.

Savings-to-Spend Ratio = Estimated Cost Savings / AI Spend, with division-by-zero protection.

Total Impact Score = ROI × (1 + Automation Rate/100) × (1 + AI Projects/10).

These measures are analytical constructs used in this project to combine financial, operational and adoption-related information. They should be interpreted as project-level analytical indicators rather than standardized industry accounting measures.

2.3 Technology Stack

Python, Pandas, NumPy, Matplotlib and Seaborn are used for data preparation and analysis. The project also contains SQL queries for MySQL, a database named ai_automation_banking, and a table named ai_automation_spend. The Python database script uses MySQL Connector/Python.

3. Exploratory Data Analysis

Exploratory analysis was used to identify changes in spending over time and differences across regions, countries, technologies, banks and AI use cases. The project notebook also examines correlations among financial and operational measures.

Recorded AI spending increased from USD 25.00 million in 2020 to USD 123.51 million in 2025.

4. Regional and Country Analysis

The dataset represents four regions: Europe, Asia-Pacific, North America and the Middle East. Regional aggregation shows that Europe has the largest recorded AI spending in this project dataset.

Europe records USD 157.22 million, followed by Asia-Pacific at USD 148.43 million and North America at USD 135.08 million. The Middle East contributes USD 106.53 million.

Canada has the highest recorded country-level spending at USD 57.12 million, narrowly followed by the USA at USD 56.98 million. These figures describe only the records in the project dataset and should not be interpreted as official national AI investment statistics.

5. Technology and Use-Case Analysis

Four technology categories are present: RPA + AI, Generative AI, Machine Learning and AI Automation. The spending distribution provides a view of where banking AI investment is concentrated within the sample.

RPA + AI has the highest recorded spending at USD 141.61 million. Generative AI has the highest average ROI in the project at 2.21x.

Back-office Workflow has the highest aggregate number of AI projects in the dataset, with 555 projects. Other use cases include document processing, customer service chatbot, credit risk and underwriting, AML monitoring, code generation, IT operations and marketing personalization.

6. Financial and Operational Impact

The project combines financial and operational indicators to assess the recorded impact of AI adoption. The aggregate results are based on the 72 records in the cleaned analytical dataset.

The estimated savings exceed recorded spending by USD 22.25 million in aggregate. This produces a positive project-level net benefit, although individual records vary substantially. Therefore, the aggregate result should not be read as a guarantee that every AI initiative is financially positive.

Production has the highest average automation rate at 70.00%, while the project also contains Pilot and Scaling maturity records. The maturity analysis is descriptive; it does not establish that maturity alone causes a higher automation rate.

7. Relationship and Correlation Analysis

Correlation analysis was used to examine linear relationships among AI spending, estimated cost savings, ROI, automation rate, FTE hours saved, project count, net benefit and savings-to-spend ratio. Correlation is used here as a descriptive analytical measure and should not be interpreted as proof of causation.

The strongest relationships in the matrix should be interpreted in the context of how the variables are constructed. For example, Net Benefit is mathematically derived from spending and estimated savings, while Savings-to-Spend Ratio also uses spending and savings. Consequently, strong correlations involving these derived variables are partly expected.

8. SQL and Database Analysis

The project includes a MySQL implementation for structured analysis. The SQL script creates the ai_automation_banking database and the ai_automation_spend table. The table stores the 13 core source fields and uses an auto-incrementing identifier.

The SQL analysis covers total record count, total AI spending, estimated cost savings, average ROI, average automation rate, total AI projects, country-wise, region-wise and year-wise spending, bank-wise spending, AI use-case spending, technology-wise analysis, maturity-wise analysis and top ROI projects.

The project contains a database_connection.py script using MySQL Connector/Python. The connection targets a local MySQL instance and the ai_automation_banking database.

9. Key Findings and Discussion

The project indicates a clear upward pattern in recorded AI spending across the available years. Annual spending rises from USD 25.00 million in 2020 to USD 123.51 million in 2025. This represents growth within the project sample, not a market-wide estimate.

At the aggregate level, AI spending is USD 547.26 million while estimated savings are USD 569.51 million. The positive net benefit of USD 22.25 million suggests that the sample, taken as a whole, contains slightly more estimated savings than recorded spending.

Technology-level results show that RPA + AI has the highest recorded spending, whereas Generative AI has the highest average ROI. This difference illustrates why investment volume and return efficiency should be evaluated separately.

Geographically, Europe is the highest-spending region, while Canada is the highest-spending country in the dataset. The results reflect the project records and not the complete banking sectors of those regions.

Operationally, the records account for 34,684 monthly FTE hours saved and an average automation rate of 61.57%. Back-office Workflow has the largest aggregate project count, indicating a notable concentration of automation activity in internal operational processes.

9.1 Practical Implications

- Banking organizations can compare investment volume with both financial and operational outcomes.
- ROI alone may not be sufficient; automation and FTE savings add an operational perspective.
- Technology categories can differ in spending intensity and average return.
- Maturity-based reporting can help distinguish pilot initiatives from scaled or production initiatives.
- Derived metrics can support prioritization, but should be interpreted together with the underlying fields.

10. Limitations

- The dataset contains 72 records and is not a complete representation of all banking AI activity.
- The recorded spending and estimated savings depend on the values present in the project dataset.
- Estimated cost savings are not independently audited financial statements.
- The project mainly performs descriptive and exploratory analysis; it does not establish causality.
- The composite Total Impact Score is a project-defined analytical metric and is not a standard banking KPI.
- External variables such as market conditions, regulatory changes and macroeconomic factors are not represented.
- Country and regional totals should not be interpreted as official national investment statistics.

11. Future Scope

- Expand the dataset with more banks, countries and years.
- Add time-series forecasting of AI spending, savings and automation.
- Build predictive models for ROI or cost-saving outcomes.
- Add bank-level benchmarking and peer-group comparison.
- Integrate a live or scheduled data-refresh process.
- Connect the cleaned dataset directly to a reporting/dashboard layer.
- Add additional risk, compliance and regulatory indicators for banking AI.
- Track project-level outcomes after implementation to compare estimates with realized savings.

12. Conclusion

This research paper presents an end-to-end analysis of AI and automation spending in the banking sector using the supplied project dataset. The work combines data cleaning, feature engineering, exploratory analysis, SQL aggregation and business interpretation. The analysis covers spending, estimated savings, ROI, automation, FTE hours saved, project adoption, maturity, geography, technology and use case.

The project records USD 547.26 million of AI spending and USD 569.51 million of estimated savings, with a net benefit of USD 22.25 million. The average ROI is 1.96x, average automation is 61.57%, and the dataset represents 5,099 AI projects. These results demonstrate how a structured data analytics workflow can convert operational AI investment records into measurable business insights.

Overall, the project provides a practical framework for evaluating AI adoption in banking. Its main value is not a single metric, but the combination of financial, operational and adoption measures that can help decision-makers understand where AI investment is occurring and how those investments are represented in the available data.

13. References and Project Artifacts

Python Software Foundation. Python documentation.

Pandas documentation – data manipulation and analysis.

NumPy documentation – numerical computing.

Matplotlib documentation – data visualization.

Seaborn documentation – statistical visualization.

MySQL documentation – relational database and SQL reference.

MySQL Connector/Python documentation – Python database connectivity.

Project raw dataset: AI_Automation_Banking_Spend_Raw_Data.csv.

Project cleaned dataset: AI_Automation_Spend_Cleaned_Data.csv.

Project analysis notebooks: AI_Automation_Spend_in_Banking_Sector_Analysis.ipynb and AI_Banking_Investment_Analysis.ipynb.

Project SQL script: AI_Automation_Banking_SQL_Queries.sql.

Project database connection script: database_connection.py.

Appendix A – Main Analytical KPIs

Metric	Result
Total AI Spend	USD 547.26 million
Estimated Cost Savings	USD 569.51 million
Net Benefit	USD 22.25 million
Total AI Projects	5,099
Average ROI	1.96x
Average Automation Rate	61.57%
Monthly FTE Hours Saved	34,684

Appendix B – Main Analytical Variables

Variable	Meaning
AI_Spend_USD_Million	Recorded AI investment amount in USD million.
Estimated_Cost_Savings_USD_Million	Estimated cost savings associated with the record.
ROI_Multiple	Recorded return-on-investment multiple.
Automation_Rate_Percent	Recorded percentage of process automation.
FTE_Hours_Saved_Monthly	Monthly FTE hours saved.
AI_Projects	Number of AI projects represented by the record.
Maturity	AI initiative maturity category.
Net_Benefit_USD_Million	Estimated savings minus AI spend.

Savings_to_Spend_Ratio	Estimated savings divided by AI spend.
Total_Impact_Score	Project-defined combined impact indicator.